FIRST SEMESTER 2024-25
COURSE HANDOUT

Date: 30.07.2024

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No : CE F231
Course Title : Fluid Mechanics
Instructor-in-Charge : Dr. Selva Balaji M (1201)
Tutorial Instructors : Dr. Selva Balaji M and Mr. Harishvar J (1201 and 1205)

1. Course Description:

Fluid Mechanics is an introductory course that covers the fundamental aspects of fluid mechanics, especially the classical and well-established laws. This course emphasizes understanding these governing laws and their widespread engineering applications in fluid related problems. The contents of this course will enable the students to tackle real-life fluid mechanics problems and provide a broad overview of the subject. Also, this course will be helpful to students as a prerequisite for further advanced courses of fluid mechanics at undergraduate and graduate levels.

2. Scope and Objective of the Course:

- Introduction to different fluid properties and their engineering applications
- Learn about pressure exerted by fluids and pressure measurement techniques
- Estimate hydrostatic forces on surfaces
- Learn about buoyancy and evaluate stability of submerged and floating bodies
- Understand concepts of fluid kinematics, methods of describing fluid flow, types of fluid flow, control mass and control volume, rate of flow, continuity equation, potential and stream functions
- Study dynamics of fluid flow, Euler's equation, Bernoulli's equation and its applications, Navier Stokes equation, and pipe bend problems
- Compute fluid discharge through orifices and mouthpieces, and flow over weirs and notches
- Analyze flow of viscous fluids through pipes and plates, and compute head loss due to friction
- Compute different types of energy losses and determine hydraulic gradient and total energy line for flow through pipes. Become adept at solving problems pertaining to syphon, flow through pipes in series, parallel and branched pipes, power transmission through pipes, and flow through nozzles
- Perceive the concepts of dimensional analysis and similitude

3. Textbook:

TB1: Munson, B.R., Okiishi, T.H., Huebsch, W.W., Rothmayer, A.P., 2009. *Fundamentals of fluid mechanics*, 6th edition, Wiley India Pvt. Ltd, New Delhi.

TB2: Modi, M. and Seth, H., *Fluid mechanics and hydraulics*. Standard Book House, New Delhi.

4. Reference Book:

RB1: Som, S.K. and Biswas, G., 2004. *Introduction to Fluid Mechanics and Fluid Machines,* 2nd edition, Tata McGraw Hill Education, Noida, India.
RB1: Streeter, V.L. and Wylie, E.B., 2017. *Fluid Mechanics*, 9th edition, McGraw Hill Education, Noida, India.
RB2: Fox, R.W. and McDonald, A.T., *Introduction to Fluid Mechanics*, John Wiley & Sons Inc., Singapore, Seventh Edition, 2011.

5. Course Plan:

Module No.	Lecture Session*	Reference	Learning outcomes
1	Introduction and Properties of fluids: L1-L3	TB1 – Ch 1 TB2 – Ch 1	Learn about different fluid properties and their engineering applications
2	Fluid pressure and its measurement: L4-L5	TB1 – Ch 2, TB2 – Ch 2	Learn about pressure exerted by fluids and pressure measurement techniques
3	Hydrostatic forces on surfaces: L6-L9	TB1 – Ch 2, TB2 – Ch 3	Procedure to estimate hydrostatic forces on surfaces
4	Buoyancy and floatation: L10-L12	TB1 – Ch 2, TB2 – Ch 4	Learn about buoyancy and evaluate stability of submerged and floating bodies
5	Kinematics of flow: L13-L19	TB1 – Ch 3, Ch 4 TB2 – Ch 5, Ch 6	Understand concepts of fluid kinematics, methods of describing fluid flow, types of fluid flow, control mass and control volume, rate of flow, continuity equation, potential and stream functions
6	Dynamics of fluid flow: L20-L25	TB1 – Ch 5, Ch 6, Ch 7 TB2 – Ch 7, Ch 8	Learn about dynamics of fluid flow, Euler's equation, Bernoulli's equation and its applications, Navier Stokes equation, and pipe bend problems
7	Flow measurement: Orifices, mouthpieces, notches, and weirs: L26-L30	TB1 – Ch 8, TB2 – Ch 9, Ch 10	Compute fluid discharge through orifices, mouthpieces, notches, and weirs
8	Viscous flow: L31-L33	TB1 – Ch 8, TB2 – Ch 11	Analyze flow of viscous fluids through pipes and plates, and compute head loss due to friction in viscous flow

9	Flow through pipes: L34-L36	TB1 – Ch 8 TB2 – Ch 12, Ch 13, Ch 14, Lecture PPT	Compute different types of energy losses and determine hydraulic gradient and total energy line for flow through pipes. Become adept at solving problems pertaining to syphon, flow through pipes in series, parallel and branched pipes, power transmission through pipes, and flow through nozzles
10	Dimensional and model analyses: L37-L38	TB1 – Ch 7 TB2 – Ch 17	Perceive and apply the concepts of dimensional analysis and similitude

***Tentative**

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Evaluative Tutorials	45 mins.	25%	Will be notified at least one week before each evaluative tutorial	Close Book/Open Book
Mid-Semester Test	90 mins.	30%	As announced in the Timetable	Close Book/Open Book
Comprehensive Examination	180 mins.	45%	As announced in the Timetable	Close Book/Open Book

7. Chamber Consultation Hour: Wednesday 4 PM – 5 PM (Please get an appointment through e-mail)

8. Notices: All announcements will be communicated via email/google classroom page

9. Make-up Policy: No make-up exams will be conducted for evaluative tutorials. Make-up request for Mid-Semester / Comprehensive examination must be notified before the respective examinations and decision will be made as per institute norms. Make-up requests made on medical grounds may not be approved unless recommended by CMO BITS Pilani (please refer to https://academic.bits-pilani.ac.in/Faculty/Download/Notice_For_Students_Requests_Nn_Medical_Background_10_March_2020.pdf)

10. Note (if any): Office – 1212-H (FD1)

Instructor-in-charge
CE F231